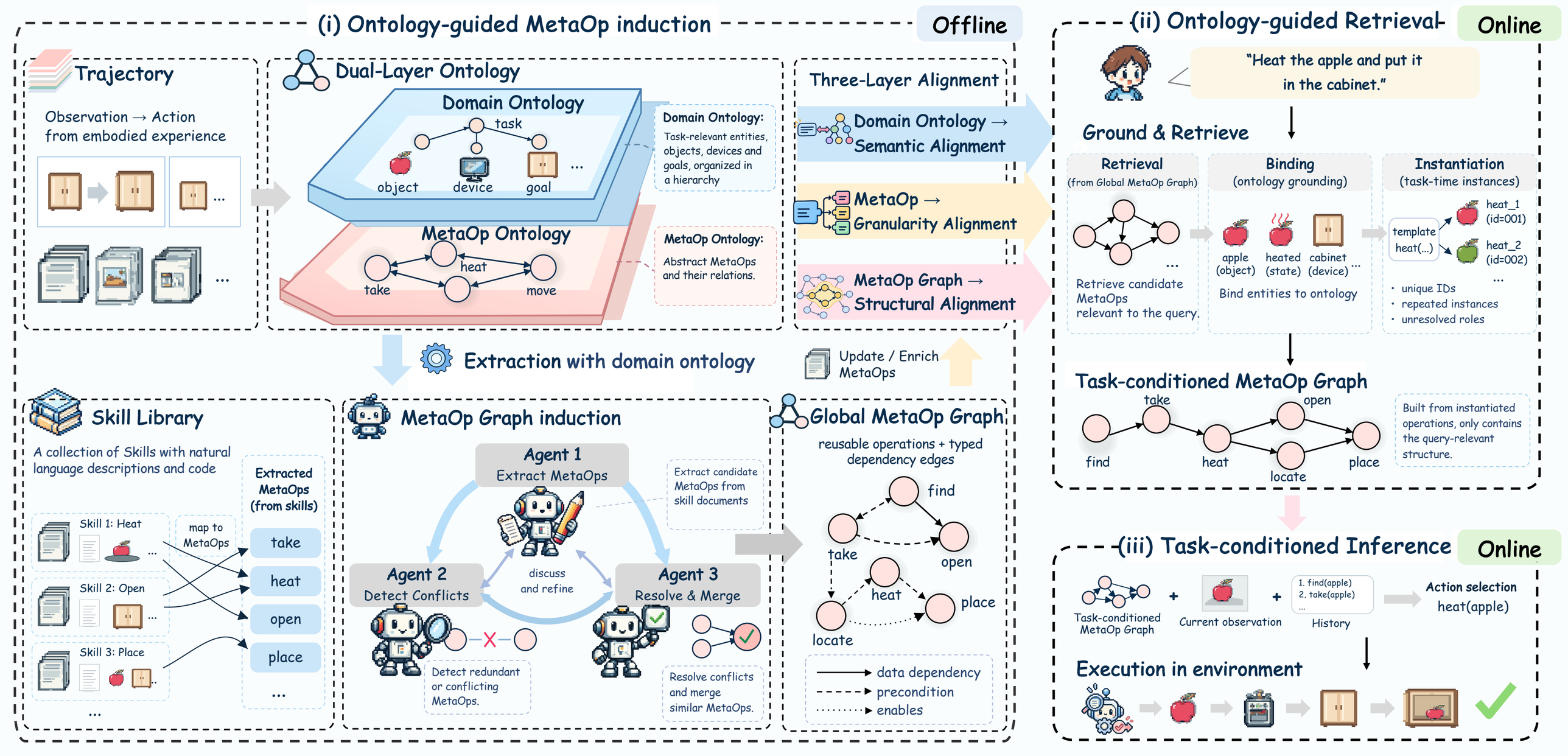

(i) Ontology-guided MetaOp induction
Offline
Trajectory
Observation → Action
from embodied experience
Dual-Layer Ontology
Domain Ontology
task
object
device
goal
Domain Ontology:
Task-relevant entities, objects, devices and goals, organized in a hierarchy
MetaOp Ontology
heat
take
move
MetaOp Ontology:
Abstract MetaOps and their relations.
Three-Layer Alignment
Domain Ontology → Semantic Alignment
MetaOp → Granularity Alignment
MetaOp Graph → Structural Alignment
Extraction with domain ontology
Update / Enrich MetaOps
Skill Library
A collection of Skills with natural language descriptions and code
Skill 1: Heat
Skill 2: Open
Skill 3: Place
map to MetaOps
Extracted MetaOps (from skills)
take
heat
open
place
MetaOp Graph induction
Agent 1
Extract MetaOps
Extract candidate MetaOps from skill documents
Agent 2
Detect Conflicts
discuss and refine
Agent 3
Resolve & Merge
Detect redundant or conflicting MetaOps.
Resolve conflicts and merge similar MetaOps.
Global MetaOp Graph
reusable operations + typed dependency edges
find
take
open
heat
place
locate
data dependency
precondition
enables
(ii) Ontology-guided Retrieval
Online
"Heat the apple and put it in the cabinet."
Ground & Retrieve
Retrieval
(from Global MetaOp Graph)
Retrieve candidate MetaOps relevant to the query.
Binding
(ontology grounding)
apple (object)
heated (state)
cabinet (device)
Bind entities to ontology
Instantiation
(task-time instances)
template heat(...)
heat_1 (id=001)
heat_2 (id=002)
unique IDs
repeated instances
unresolved roles
Task-conditioned MetaOp Graph
take
open
find
heat
locate
place
Built from instantiated operations, only contains the query-relevant structure.
(iii) Task-conditioned Inference
Online
Task-conditioned MetaOp Graph
Current observation
1. find(apple)
2. take(apple)
...
History
Action selection
heat(apple)
Execution in environment